

Notation Guide

This guide collects the notation used across the regression portion of the text. It is not a replacement for the chapter explanations; it is a quick reference for keeping the object types straight.

Object Types

Notation	Object type	How to read it
x, y, b, p, n	scalars	Single numerical values or indices.
X, Y, ϵ	random variables	Population-level or model-level random quantities.
$\mathbf{x}, \mathbf{y}, \mathbf{e}$	vectors	Observed sample-level vectors, usually in $\mathbb{R}^n$.
$\mathbf{X}, \mathbf{H}, \mathbf{I}_n$	matrices	Data, projection, or identity matrices.
$\boldsymbol{\beta}, \boldsymbol{\epsilon}$	parameter/error vectors	Coefficient and error vectors in regression models.
$\hat{}$	estimate or fitted quantity	A hat marks something estimated from data or predicted by a fitted model.

Data and Statistical Learning Notation

For a single observation, $\mathbf{x} = (x_1, \dots, x_p)^T$ is the vector of p input values and Y is the response. The general supervised-learning model is

$$Y = f(\mathbf{X}) + \epsilon,$$

where f is the systematic relationship and ϵ is the unexplained error.

For a data set with n observations, i indexes observations and j indexes variables:

$$x_{ij} = \text{value of predictor } j \text{ for observation } i.$$

The observed response vector is

$$\mathbf{y} = (y_1, \dots, y_n)^T.$$

The notation $\mathbf{x}_j$ means the full observed column of predictor j . The notation $\mathbf{x}_i^T$ means the row of predictors for observation i .

Regression Notation

In the regression chapters, the design matrix $\mathbf{X}$ includes an intercept column unless stated otherwise. With p predictors and an intercept,

$$\mathbf{X} \in \mathbb{R}^{n \times (p+1)}, \quad \boldsymbol{\beta} \in \mathbb{R}^{p+1}.$$

The fitted values and residuals are

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}, \quad \mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}.$$

The true error vector is $\boldsymbol{\epsilon}$; the residual vector is $\mathbf{e}$. Errors are part of the data-generating model and are not observed. Residuals are computed after fitting a model.

The ordinary least squares estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y},$$

when $\mathbf{X}$ has full column rank. The hat matrix is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T, \quad \hat{\mathbf{y}} = \mathbf{H}\mathbf{y}.$$

The residual-maker matrix is $\mathbf{I}_n - \mathbf{H}$.

Observed Vector Versus Random Vector

Most chapters use $\mathbf{y}$ for the observed response vector from the data set in hand. Chapter ?? temporarily uses $\mathbf{Y}$ because it studies the fixed-design theoretical model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where $\mathbf{X}$ is treated as fixed and $\mathbf{Y}$ varies randomly through $\boldsymbol{\epsilon}$. After that theory chapter, the text returns to $\mathbf{y}$ when discussing a realized fitted model.

Fit, Testing, and Diagnostics

Notation	Meaning
RSS	Residual sum of squares, $\sum_i (y_i - \hat{y}_i)^2 = \ \mathbf{e}\ ^2$.
TSS	Total sum of squares, $\sum_i (y_i - \bar{y})^2$.
RegSS	Regression or explained sum of squares, often TSS – RSS when an intercept is included.
R^2	Proportion of response variation explained relative to the mean-only baseline.
RSE	Residual standard error, the estimated residual standard deviation.
h_{ii}	Leverage of observation i , the i th diagonal entry of $\mathbf{H}$.

The Big Three Regression Assumptions

The regression chapters use a common three-step assumption ladder:

1. **Mean-zero errors:** the model is centered correctly, e.g. $\mathbb{E}[\epsilon \mid \mathbf{X}] = \mathbf{0}$.
2. **Constant variance and uncorrelated errors:** $\text{Var}(\epsilon \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$.
3. **Gaussian errors:** $\epsilon \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

The stronger assumptions support stronger inferential claims, but they are not all needed for every modeling task.

Linear Models

Early chapters use “linear” in the familiar geometric sense: a line, plane, or hyperplane in the displayed predictors. Chapter ?? sharpens the definition: for least squares, the key requirement is that the model be linear in the unknown coefficients. Curved functions of the original predictors can still be linear models if they can be written as a design matrix times a coefficient vector plus error.